

assignment - static equilibrium

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:44pm

Quiz Instructions

There is a tutorial in dropbox I think, try without hints

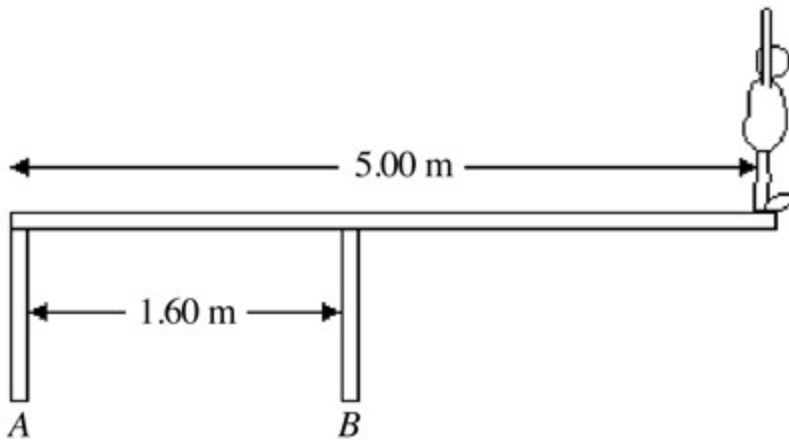


Question 1 11.2 pts

An 82.0 kg diver stands at the edge of a light 5.00-m diving board, which is supported by two narrow pillars 1.60 m apart, as shown in the figure. Find the magnitude and direction of the force exerted on the diving board

(a) by pillar A.

(b) by pillar B.



hint: 1) Find the force from B first (upward). Place the pivot at A. Use $\sum (CCW) = \sum (CW)$

2) Find the force from A second (downward). Place the pivot at B. Use $\sum (CCW) = \sum (CW)$

see slide

☐

150N, 250N

☐

1.71kN, 2.51kN

☐

2.5kN, 1.5kN

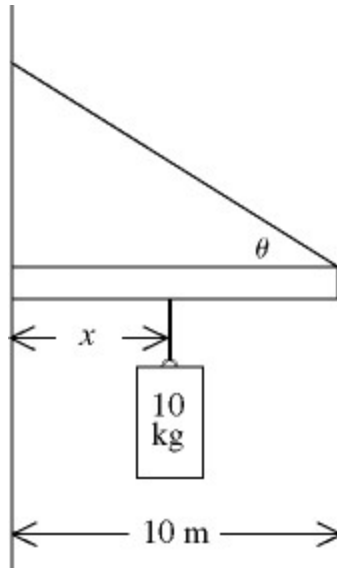
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1.5kN, 2kN



Question 2 11.2 pts

In the figure, a 10.0-m long bar is attached by a frictionless hinge to a wall and held horizontal by a rope that makes an angle θ of 53° with the bar. The bar is uniform and weighs 39.9 N. How far from the hinge should a 10.0-kg mass be suspended for the tension T in the rope to be 125 N? (about)



hint: Remember that the weight of an uniform bar is located at its center. Remember that weight (N) is mass $\times$ 9.8. Take the pivot at the contact point between the bar and the wall. Use: Sum (clockwise torques) = Sum (counter clockwise torques).



5m from hinge



8m from hinge

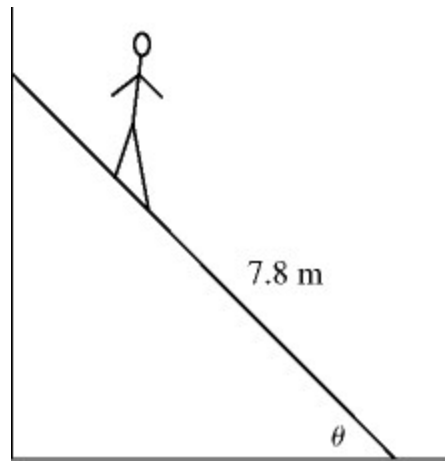


3m from hinge



Question 3 11.2 pts

In the figure, a uniform ladder 12 meters long rests against a vertical frictionless wall. The ladder weighs 400 N and makes an angle θ of 79° with the floor. A man weighing 790 N climbs slowly up the ladder. When he has climbed to a point that is 7.8 m from the base of the ladder, the ladder starts to slip. What is the coefficient of static friction between the floor and the ladder?



hint: see slides. you have weight of man, weight of ladder (down), normal from ground N_1 and normal from the wall N_2 , and the frictional force. You can place the pivot at the contact point between ground and ladder. You are left with torques from N_2 and the 2 weights. $CCW(\text{weights}) = CW(N_2)$. To decide if it's sine or cosine. It's easier if you use the equation $\text{torque} = \text{force} \times \text{lever arm}$. lever arm is the perpendicular distance between the axis and the line of action of the force.

☐

.12

☐

0.1

☐

0.12

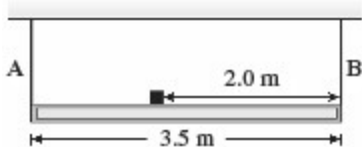
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0.11

☒

Question 4 11.2 pts

A uniform 1200-N piece of medical apparatus that is 3.5 m long is suspended horizontally by two vertical wires at its ends. A small but dense 550-N weight is placed on the apparatus 2.0 m from one end, as shown in the figure. What are the tensions, A and B , in the two wires?



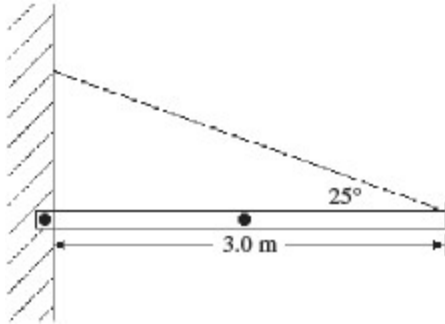
hint: Consider forces acting on the rod. Its weight, the weight of the mass, 2 tensions. Up=down. Then take the pivot at A or B. You have your 2 equations. Easy

☐
 $A = 9000 \text{ N}, B = 8200 \text{ N}$
☐
 $A = 840 \text{ N}, B = 910 \text{ N}$
☐
 $A = 910 \text{ N}, B = 840 \text{ N}$

 $A = 880 \text{ N}, B = 880 \text{ N}$  $A = 8200 \text{ N}, B = 9000 \text{ N}$ 

Question 5 11.2 pts

A store's sign has a mass of 20 kg and is 3.0 m long. It is uniform, so its center of gravity is at the center of the sign. It is supported horizontally by a small loose bolt attached to the wall at one end and by a wire at the other end, as shown in the figure. What is the magnitude of the *net* force that the bolt exerts on the sign?



hint: we did it in class.



120 N



230 N



460 N



200 N

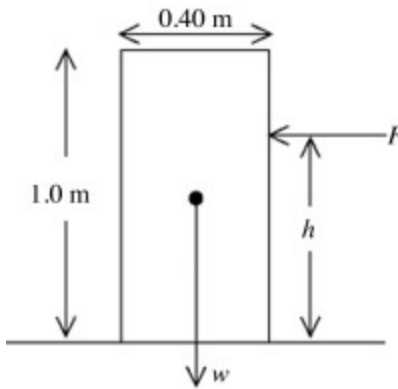


300 N



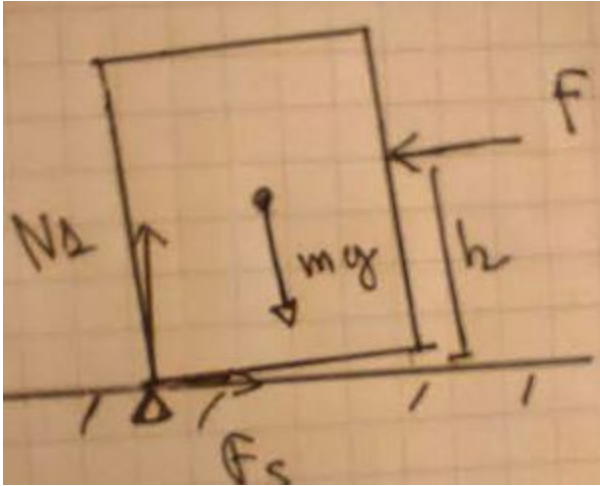
Question 6 11.2 pts

In the figure, a uniform rectangular crate 0.40 m wide and 1.0 m tall rests on a horizontal surface. The crate weighs 930 N, and its center of gravity is at its geometric center. A horizontal force F is applied at a distance h above the floor. If $h = 0.61 \text{ m}$, what minimum value of F is required to make the crate start to tip over? Static friction is large enough that the crate does not start to slide. round to nearest one



hint: Try on your own. Consider the crate as it starts to tip over and be in contact with the ground only with its left corner. Use 9.8. round to the nearest one.

Take the pivot at the left corner. only 2 torques are involved. weight and force. Use the equation with the lever arm. torque = force $\times$ lever arm. Its easier. Extend the forces (lines of action) and find the perpendicular distance to the line of action of the force (projection of the axis). easy


☐

100N

☐

200N

☐

305N

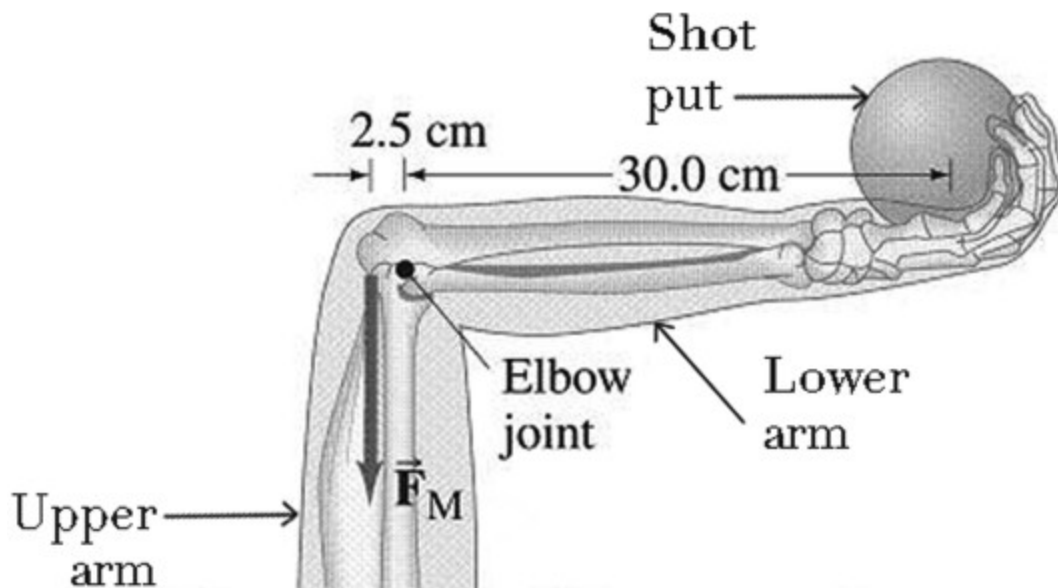
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310N

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Question 7 11.2 pts

In the figure, the horizontal lower arm has a mass of 2.8 kg and its center of gravity is 12 cm from the elbow joint pivot. How much force F_M must the vertical extensor muscle in the upper arm exert on the lower arm to hold a 7.5 kg shot put?



hint: we did it in class. You don't need to convert cm to m. The weight of the lower arm is applied at its center of mass (half)

- ☐ 500 N
- ☐ 1500 N
- ☐ 750 N
- ☐ 1000 N
- ☐ 100 N
- ☒

Question 8 11.2 pts

A 20.0-kg uniform plank is supported by the floor at one end and by a vertical rope at the other as shown in the figure. A 50.0-kg mass person stands on the plank a distance three-fourths of the length plank from the end on the floor.. No friction involved



(a) What is the tension in the rope?

(b) What is the magnitude of the force that the floor exerts on the plank?

hint: important problem.

1) first draw the forces, tension T is up, normal force N from ground is up. weight of person ($L/2$) and

weight of plank($0.75 L$ from gnd). are up. angle with ground is a .

2) Use $up = down$ and you find $T + N = \dots$ (remember weight is $mass \times 9.8$) Equ1

3) Now take the pivot at the contact between the plank and the ground.

The weights produce CW torques and $T_{\perp}$ ($T \cos(a)$) produces a CCW torque
 $sum(CCW) = sum(CW)$. The distances involved are $L/2$, L , $0.75L$.

You get $T \cos(a) = \dots$ Equ2 (note no torque by N since it is at the pivot)

4) take the pivot at the connection between rope and plank. Same as before.

distances involved are L , $0.5L$, $0.25L$. You get $N \cos(a) = \dots$ Equ3.

5) divide Equation 2 by equation 3. $\cos(a)$ disappears and you should get $T = 2.1$ (normal)
 plug into Equ 1 . Et voila

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466N, 220N

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200N, 420N

☐

150N, 250N

☐

320N, 450N

⋮

Question 9 11.2 pts

A 3.00-kg ball rests in a frictionless groove as shown in the figure.



(a) What is the magnitude of the force that the left side of the groove exerts on the ball?

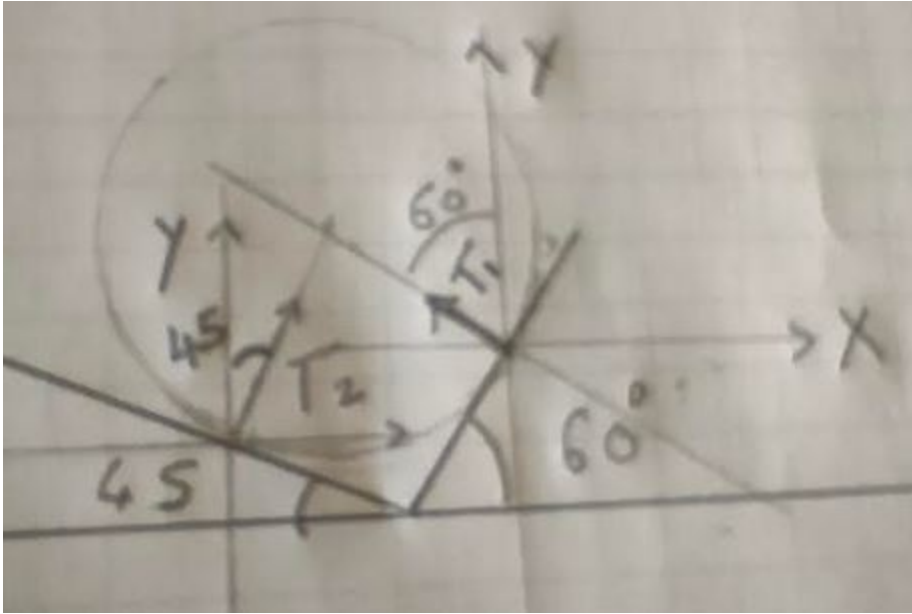
(b) What is the magnitude of the force that the right side of the groove exerts on the ball?

hint: This is not as hard as it seems. Get graph paper, a ruler, compass. exaggerate the angles (don't peek below. Try first on your own.

You don't need torques. Just $up = down$ and $left = right$.

You get 2 equations with 2 unknown (T_1 and T_2) . Use the TI you find the answers so

you get the same as mine.

☐

20.5, 25.5

☐

26.4N, 21.5N

☐

18.5N, 25.1N

☐

21.3N, 22.1N

Not saved

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